

GIDDINGS LEE COUNTY AP, TX, USA

WMO: 722113

Lat: 30.169N

Lon: 96.980W

Elev: 148

StdP: 99.56

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-0.3	-9.8	1.7	4.4	-7.6	2.0	6.1	9.4	11.7	8.4	13.1	2.2	10	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	37.7	23.2	36.8	23.1	35.6	23.3	26.3	31.4	25.7	30.9	25.3	30.5	3.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.8	27.6	24.7	20.0	27.2	24.0	19.2	26.8	82.6	31.4	80.0	31.0	77.9	30.5	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	7.0	5.8	-5.2	39.3	2.7	1.3	-7.2	40.2	-8.8	40.9	-10.4	41.6	-12.4	42.6		
			-5.8	26.2	2.6	1.0	-7.7	26.9	-9.2	27.5	-10.7	28.0	-12.7	28.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	10.6	12.5	16.5	20.1	24.1	27.9	29.0	29.7	26.5	21.3	15.7	11.4
	DBStd	7.89	5.36	5.72	4.79	4.04	3.23	1.94	1.67	1.98	2.75	4.41	5.23	5.28
	HDD10.0	161	57	37	8	1	0	0	0	0	0	1	12	46
	HDD18.3	897	245	176	91	28	3	0	0	0	0	23	110	222
	CDD10.0	3988	76	106	211	304	436	537	589	609	495	352	183	91
	CDD18.3	1684	6	11	36	81	181	287	331	351	246	117	31	8
	CDH23.3	17846	23	48	196	598	1634	3235	3933	4385	2502	1064	193	35
	CDH26.7	8554	1	4	23	144	618	1618	2073	2452	1214	375	31	2
(14) Wind	WSAvg	2.5	2.8	3.0	3.0	3.0	2.7	2.4	2.2	2.2	1.8	2.1	2.3	2.5
(15) Precipitation	PrecAvg	1018	81	74	73	82	106	97	60	67	95	119	89	78
	PrecMax	1445	287	197	169	179	266	281	179	187	205	386	261	255
	PrecMin	550	4	5	27	14	14	1	1	9	11	0	11	24
	PrecStd	267	61	50	35	47	64	65	46	47	55	97	69	54
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	27.3	28.7	32.3	34.5	37.9	38.2	39.4	38.0	34.1	29.1	26.3
		MCWB	17.3	17.7	19.5	20.1	22.5	22.5	22.8	24.4	22.8	22.4	20.8	18.4
	2%	DB	23.0	24.9	27.3	30.1	32.8	36.2	37.2	37.9	36.0	32.2	27.5	24.0
		MCWB	16.4	17.6	18.4	20.3	22.7	22.5	22.9	23.7	22.6	22.5	20.0	17.9
	5%	DB	21.4	22.8	25.9	27.9	32.0	35.0	36.1	37.1	34.1	30.5	25.9	22.4
		MCWB	16.5	17.4	18.3	20.1	22.4	23.1	23.2	23.6	23.0	21.9	19.2	17.8
	10%	DB	19.9	21.2	23.8	27.0	30.2	33.4	34.9	35.9	32.6	28.8	23.7	20.8
		MCWB	15.8	16.9	18.1	19.4	22.2	23.2	23.5	23.5	22.7	20.5	18.8	17.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.7	21.5	23.6	25.7	27.2	26.7	27.0	26.0	25.2	23.0	21.6
		MCDB	21.9	23.2	25.5	27.7	30.6	32.3	33.1	30.6	30.6	28.7	26.6	23.6
	2%	WB	19.1	19.8	20.7	22.5	24.7	25.6	26.0	26.4	25.4	24.4	21.9	20.4
		MCDB	21.6	22.8	23.7	26.9	29.4	30.4	31.4	31.3	29.8	28.4	24.8	22.7
	5%	WB	18.2	18.9	20.1	21.9	24.0	25.0	25.5	25.7	24.8	23.6	21.1	19.1
		MCDB	20.6	21.9	23.2	26.2	28.5	30.0	30.4	31.4	29.4	27.8	23.6	21.6
	10%	WB	16.6	17.8	19.4	21.2	23.3	24.5	25.0	25.3	24.3	22.9	20.1	17.8
		MCDB	18.7	20.1	22.5	25.0	27.8	29.4	29.8	31.4	29.1	27.0	22.9	19.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	10.9	11.4	11.5	10.7	11.0	11.2	12.0	11.5	12.4	11.7	11.2
		MCDBR	13.2	12.3	11.9	11.9	11.6	12.6	13.0	13.4	13.2	12.6	12.4	13.0
		MCWBR	7.8	7.4	5.7	5.2	3.5	2.6	2.6	2.5	3.4	4.8	6.5	8.2
	5% WB	MCDBR	10.0	10.2	9.5	9.4	9.4	9.8	10.3	11.9	11.0	10.4	9.9	10.6
		MCWBR	7.6	7.4	6.0	4.8	3.8	3.1	2.6	2.7	3.1	5.0	6.5	8.4
(40) Clear-Sky Solar Irradiance	taub		0.329	0.340	0.364	0.407	0.457	0.437	0.453	0.443	0.420	0.374	0.367	0.323
	taud		2.489	2.471	2.406	2.301	2.207	2.312	2.302	2.318	2.335	2.447	2.400	2.518
	Ebn at Noon		905	926	921	887	841	853	838	844	853	876	849	894
	Edh at Noon		91	101	115	132	146	131	132	128	122	102	98	84
(44) All-Sky Solar Radiation	RadAvg		2.93	3.35	4.43	5.34	5.78	6.46	6.52	6.16	5.14	4.28	3.19	2.59
	RadStd		0.30	0.62	0.41	0.44	0.45	0.45	0.52	0.34	0.47	0.56	0.36	0.33

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	+88

Nomenclature: See separate page